UNITED INTERNATIONAL UNIVERSITY

Final Exam | Spring 2024

CSE 4165/CSE 465 Web Programming

Mark: 30 | Time: 2 Hours (including submission)

NAME:	**ID:**

Return the question to the invigilator at the end of the exam. Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. **[10]**

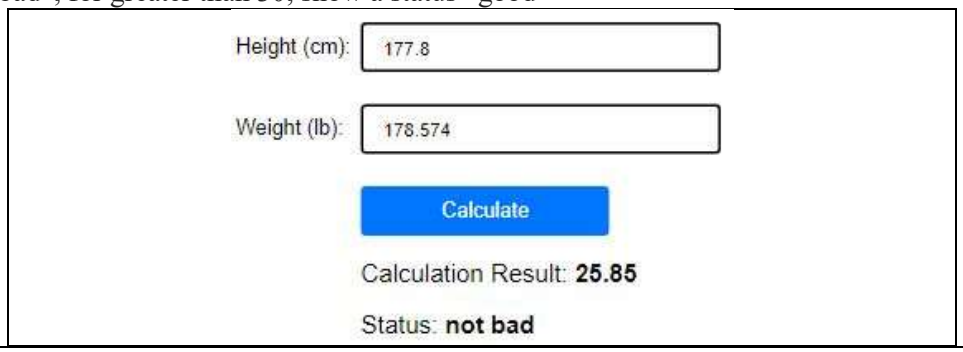
Javascript: Write a javascript program to calculate the formula given below. First, take height in centimetre(cm) and weight in pounds(lb), convert, calculate, and display results in the inner html.

Calculation Result = $\frac{\text{Weight in KG}}{(\text{Height in metre})^2}$

Formula to convert centimetre to metre: $\text{metre}=\frac{\text{centrimetre}}{100}$

Formula to convert pound to KG: $\text{KG}=\frac{\text{pound}}{2.2046}$

If the calculation result is greater than 0 and less than 20, show a status "bad", greater than 20 to 30, show a status "not bad", for greater than 30, show a status "good"



2. **[10]**

PHP: Create a cryptic sequence generator using for loop. The sequence will start with a given seed value and follow these rules:

1. The first number is the seed itself.
2. For each subsequent number in the sequence, take the previous number and add the product of its digits.
3. Repeat this process for a specified number of terms.

Now, write a PHP program that takes a seed value and the number of terms as input and generates the cryptic sequence based on the rules above. Display the sequence in a comma-separated format.

Sample Input	Sample Output
Seed value: 7 Terms: 5	7, 14, 19, 29, 40
Seed value: 4 Terms: 9	4, 8, 16, 23, 28, 38, 49, 62, 70

3. **[10]**

PHP&MYSQL: Design and write the following web application using php and all other supporting languages. The web app will implement an electricity bill calculation system. The bills will vary depending on a consumer's location, area, and quantity consumed unit. First, create an *electricity_bill* database, and then create a *bill_info* table with the following fields:

#	Name	Type
1	id	int(11)
2	location	varchar(100)
3	area	varchar(100)
4	rate_0_75	float
5	rate_76_200	float
6	rate_201_above	float

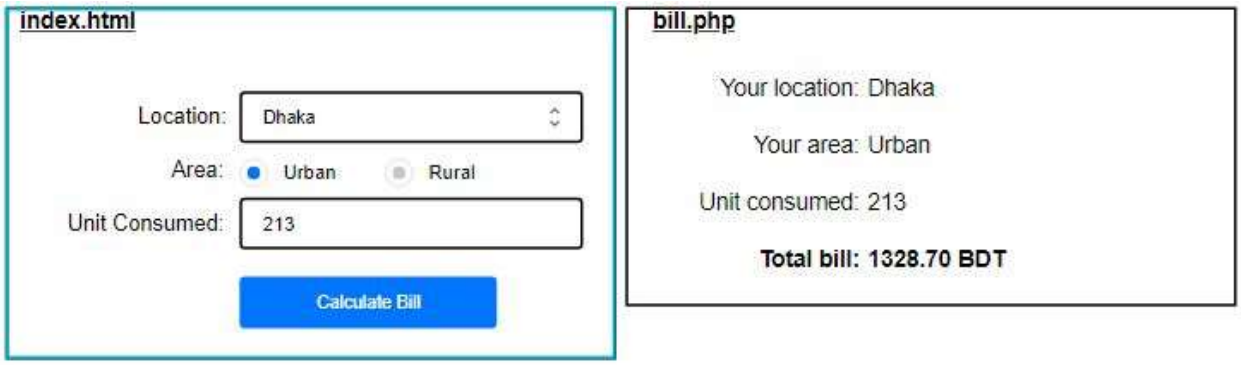
Fill the above table with the following billing/rate information:

Location	Area	Rates (Unit=BDT)
Dhaka	Urban	0-75 = 4.19 76-200 = 5.72 201-above = 6.00
	Rural	0-75 = 3.19 76-200 = 4.72 201-above = 5.00
Chittagong	Urban	0-75 = 3.50 76-200 = 4.80 201-above = 5.23
	Rural	0-75 = 2.50 76-200 = 3.80 201-above = 4.23

Tax will be applied to the bill to calculate the final Total bill. Impose tax as follows:

Location	Tax on bill
Dhaka	20%
Chittagong	15%

Now, create a *index.html* file and take all necessary input, such as location, area, and unit consumed, and then create a *bill.php* file to fetch necessary information from database based on inputs from index.html and calculate the total bill and show results as follows:



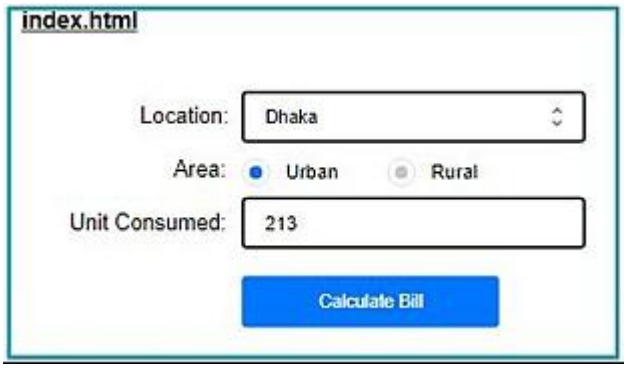
Example (MySQLi Object-oriented)

```php
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";
$conn = new mysqli($servername, $username, $password,
$dbname);
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
$sql = "INSERT INTO MyGuests (firstname, lastname,
email)
VALUES ('John', 'Doe', 'john@example.com')";
if ($conn->query($sql) === TRUE) {
  echo "New record created successfully";
} else {
  echo "Error: " . $sql . "<br>" . $conn->error;
}
$conn->close();
-----------------------------------------------------
$sql = "SELECT id, firstname, lastname FROM MyGuests";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
  // output data of each row
  while($row = $result->fetch_assoc()) {
    echo "id: " . $row["id"]. " - Name: " .
$row["firstname"]. " " . $row["lastname"]. "<br>";
  }
} else {
 echo "0 results";
}
-----------------------------------------------------
$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";
if ($conn->query($sql) === TRUE) {
  echo "Record updated successfully";
} else {
  echo "Error updating record: " . $conn->error;
}
-----------------------------------------------------
$sql = "DELETE FROM MyGuests WHERE id=3";
if ($conn->query($sql) === TRUE) {
  echo "Record deleted successfully";
} else {
  echo "Error deleting record: " . $conn->error;
}
```

United International University
CSE 4165: Web Programming
Sections: All
Final Exam | Summer 2024
Mark: 40 | Time: 2 Hours (last 10 minutes for submission only)
Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

<table>
<tr>
<td>1)</td>
<td>Calculate the electricity bill using location-specific rates. For Dhaka, charge 7.5 taka per unit in urban areas and 6.5 taka in rural areas. Chattogram rates are 7.0 taka (urban) and 6.0 taka (rural). Other locations use 6.5 taka (urban) and 5.5 taka (rural). Apply a surcharge to the total bill: no surcharge for bills under 500 taka, 5% for bills between 500-1000 taka, and 10% for bills over 1000 taka. Display the final amount, including surcharge, in an alert box, rounded to two decimal places. Implement input validation to ensure a location is selected, an area type (Urban or Rural) is chosen (can’t select both), and the "Unit Consumed" field contains a positive number. If any validation fails, display an appropriate error message to the user.

 
</td>
<td>[10]</td>
</tr>
<tr>
<td>2)</td>
<td>Karim went up a hill for 18 minutes at a speed of 20 miles per hour (km/h). Then he came back down the same hill at a speed of 60 km/h. What was his average speed for the whole trip? (Hint: It was 30 mph.). Write a PHP function that calculates the average speed for a trip when given the time spent going uphill (in minutes), the speed going uphill (in km/h), and the speed going downhill (in km/h). The function will return the average speed as a whole number (integer) in km/h.

<table>
<tr><th>Sample Input</th><th>Sample Output</th></tr>
<tr><td>calculate(18, 20, 60);
Here, 18 = time spent going uphill in minutes
20 = speed of going uphill in km/h
60 = speed of going downhill in km/h</td><td>30</td></tr>
<tr><td>calculate(30, 8, 24);</td><td>12</td></tr>
</table></td>
<td>[10]</td>
</tr>
<tr>
<td>3)</td>
<td>Write a program to check the strength of your password and display the result in your PHP file.
Strong: The password contains at least 10 characters, includes at least one uppercase letter, one lowercase letter, one digit, one special character, and ends with the symbol ‘*’.
Medium: The password contains at least 8 characters, includes at least one uppercase letter, one lowercase letter, and one digit.
Poor: The password at least 6 characters and includes at least one uppercase letter.
Very Poor: anything other than the criteria mentioned above.


</td>
<td>[10]</td>
</tr>
</table>

Sample Input	Sample Output
aFDF@F1ds*	Strong
af15f15a4f45af4asas12	Very Poor

4) A retail company wants a detailed sales analysis for January 2024 to gain deeper insights into their business performance. You are tasked with generating various reports using the sales table. The table structure is as follows: [10]

sale_id	product_id	quantity_sold	sale_date	total_price
1	107	5	2024-01-01	2500.00
2	103	3	2024-01-02	900.00
3	103	2	2024-01-02	60.00
4	107	4	2024-01-03	80.00
5	107	6	2024-01-03	90.00

Now, do the following using MYSQL with the help of PHP where needed:

1) Find the total quantity sold and the total revenue for each product.
2) Find the two days in January 2024 with the highest total revenue. Show the sale_date and the total revenue for each day.
3) Calculate the average quantity sold per product for each product_id in January 2024, but only for products that had sales on at least two different days.

Example Codes (MySQLi Object-oriented)

```
<?php
 $servername = "localhost";
 $username = "username";
 $password = "password";
 $dbname = "myDB";
 $conn = new mysqli($servername, $username, $password,
$dbname);
 if ($conn->connect_error) {
   die("Connection failed: " . $conn->connect_error);
 }
 $sql = "INSERT INTO MyGuests (firstname, lastname, email)
 VALUES ('John', 'Doe', 'john@example.com')";
 if ($conn->query($sql) === TRUE) {
   echo "New record created successfully";
 } else {
   echo "Error: " . $sql . "<br>" . $conn->error;
 }
 $conn->close();
 ----------------------------------------------------------

$sql = "SELECT id, firstname, lastname FROM MyGuests";
 $result = $conn->query($sql);

 if ($result->num_rows > 0) {
   // output data of each row
   while($row = $result->fetch_assoc()) {
     echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
$row["lastname"]. "<br>";
  }
 } else {

 echo "0 results";
 }

----------------------------------------------------------

$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";
 if ($conn->query($sql) === TRUE) {
   echo "Record updated successfully";
 } else {
   echo "Error updating record: " . $conn->error;
 }

----------------------------------------------------------

$sql = "DELETE FROM MyGuests WHERE id=3";
 if ($conn->query($sql) === TRUE) {
   echo "Record deleted successfully";
 } else {
   echo "Error deleting record: " . $conn->error;
 }
```

United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2024 Trimester: Fall

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. **[10]**

Create a JavaScript game where a user tries to guess a secret number generated by your code. Strategies are as follows:

Index.html:

Create a text filed (input type text) to take the users' guess and a submit button, below the submit button there is a label field to show the feedbacks and results.

Script.js:

1. Generate a random secret number between 500 and 5000.
2. Take the user's guess from the input field in the HTML file.
3. Inside a function:
   a) Compare the user's guess to the secret number.
   b) Display "Too high!" or "Too low!" in the feedback area of your web page.
   c) If the guess is correct, display "Correct!" and optionally disable the input field to prevent further guesses.
   d) Keep track of the number of guesses the user has made.
   e) If the user runs out of guesses (after 5 attempts), display "Out of guesses!" and optionally disable the input field.

Sample Input	**Sample Output**	
User Guesses (from html)	**Output/Feedback** (on html file using JS file)	**Result**
50, 25, 35, 38, 37	*(suppose random number:37)* Too high!, Too low!, Too low!, Too high!, Correct!	Correct!
10, 50, 90, 80, 85	*(suppose random number:82)* Too low!, Too low!, Too high!, Too low!, Too high!	Out of guesses!
1, 5, 10, 20, 15	*(suppose random number:15)* Too low!, Too low!, Too low!, Too high!, Correct!	Correct!

2. **[10]**

PHP:

It's the end of the Web Programming course, and we want to celebrate it by having a pizza party. You need to figure out how many pizzas to order and how many slices will be left over. Write a PHP function that takes three integers as input: the number of students in the class, the number of slices each student wants, and the number of slices each pizza has. The function should calculate the minimum number of pizzas you need to order to ensure every student gets enough pizza (you must order whole pizzas; you cannot order fractions of pizzas). The function should also calculate the number of pizza slices that will be left over after all the students have had their fill. Moreover, if the price of each pizza is 1050 BDT (irrespective of size), how much money was wasted based on the leftover slices?

Sample Input (from html file text fields)			**Sample Output** (print on PHP file)		
Number of Students:	Slices per Student:	Slices per Pizza:	Total Pizzas:	Leftover Slices:	Wasted Money (BDT):
10	3	8	4	2	262.5
7	2	6	3	4	700
12	2	4	6	0	0

3. **PHP & MYSQL:** **[10]**

Create a dataset named ‘uiuweb_final’ and a table named ‘student_final’. The student table has the following structure and values:

StudentID	StudentName	CourseID	CourseTitle	Grade	LetterGrade
1	Karim Uddin	101	Web Programming	85	B
2	Rahim Ahmed	101	Web Programming	92	A
3	Jashim Hossain	102	Project Management	78	C
4	Jasica Ahmed	101	Web Programming	65	D
5	Faria Karim	102	Project Management	95	A
6	Niassoh Dihan	103	System Analysis and Design	80	B

Now, do the following:

1. Show the total number of students who received each letter grade (A, B, C, D) across all courses.
2. If a student has a grade below 75 and their current letter grade is not 'D', change their letter grade to 'C'.
3. If a student has a grade greater than 80, add 5 bonus points to their grade, but only if the resulting grade is less than or equal to 90.
4. For each course, display the course titles and the number of students enrolled in that course, sorted by the number of students (most popular first).

United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2025 Trimester: Spring

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. **JAVASCRIPT:** Create a JavaScript application where users can test their password strength and get real-time feedback. **[10]**

index.html:

Create a password input field (input type password), a "Check Strength" button, and below that a label field to display strength feedback and scoring results.

script.js:

1. Generate a target strength score of 100 points for a "Very Strong" password.
2. Take the user's password input from the HTML file.
3. Inside a function:
   - Calculate password strength based on these criteria:
     - Length: +10 points for every 2 characters (minimum 6 characters required)
     - Uppercase letters: +15 points if present
     - Lowercase letters: +15 points if present
     - Numbers: +20 points if present
     - Special characters (!@#$%^&*): +25 points if present
   - Display strength levels: "Very Weak" (0-30), "Weak" (31-50), "Medium" (51-70), "Strong" (71-90), "Very Strong" (91+)
   - If password reaches 100+ points, display "Perfect Password!" and show a success message
   - Keep track of how many attempts the user has made
   - If the user makes more than 8 attempts without reaching "Strong" level, display "Need practice!" and suggest password tips

Sample Input	Sample Output
abc	Very Weak
hello123	Weak
Strong@Pass	Strong
Perfect@Pass123	Perfect Password!

2. **PHP:** It's time to organize the annual UIU Tech Fest, and you need to calculate the budget for venue booking. Write a PHP function that takes three integers as input: the number of expected attendees, the cost per person for venue arrangement, and the maximum capacity of available venues. The function should calculate the minimum number of venues needed to accommodate all attendees (you must book complete venues; you cannot book partial venues). The function should also calculate how many empty seats will remain after all attendees are seated. Moreover, if each venue costs 15,000 BDT regardless of occupancy, how much money is wasted on empty seats? **[10]**

Sample Input (from html file text fields)			Sample Output (print on PHP file)		
Attendees	Cost per Person	Venue Capacity	Total Venues	Empty Seats	Wasted Money (BDT)
250	120	80	4	70	8,400

180	150	100	2	20	3,000
300	100	75	4	0	0

3. **PHP & MYSQL:** Create a database named 'uiutech_final' and a table named 'employee_final'. The employee table has the following structure and values: **[10]**

EmployeeID	EmployeeName	DepartmentID	DepartmentName	Salary	PerformanceRating
1	Arif Rahman	201	Software Development	45000	B
2	Marium Khan	201	Software Development	52000	A
3	Sabbir Hossain	202	Quality Assurance	38000	C
4	Samira Begum	203	UI/UX Design	42000	B

Now, do the following (write full code, not just SQL queries):

1. Show the total number of employees who received each performance rating (A, B, C, D) across all departments.
2. If an employee has a salary below 40,000 BDT and their current performance rating is not 'D', change their performance rating to 'C'.
3. If an employee has a salary greater than 50,000 BDT, add a 5,000 BDT bonus to their salary, but only if the resulting salary is less than or equal to 60,000 BDT.
4. For each department, display the department names and the number of employees working in that department, sorted by the number of employees (largest department first).

Sample SQL Query

```
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";
$conn = new mysqli($servername, $username, $password,
$dbname);
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
$sql = "SELECT id, firstname, lastname FROM MyGuests";
$result = $conn->query($sql);
if ($result->num_rows > 0) {
  while($row = $result->fetch_assoc()) {
    echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
$row["lastname"]. "<br>";
  }
} else {
  echo "0 results";
}
$conn->close();
```

```
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";
$conn = new mysqli($servername, $username, $password,
$dbname);
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";
if ($conn->query($sql) === TRUE) {
  echo "Record updated successfully";
} else {
  echo "Error updating record: " . $conn->error;
}
$conn->close();
```

United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2025 Trimester: Summer

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. **[10]**

Create a JavaScript application where users can track their daily calorie intake and get real-time feedback on their eating habits.

Index.html:

Create a number input field to enter calories consumed per meal, a **"Add Calories"** button, and below that a label field to display total calories, daily goal progress, and feedback messages

Script.js:

1. Set a **daily calorie goal** of **2000 calories**.
2. Take the user's calorie input from the HTML file.
3. Inside a function:
   a) Add the entered calories to a running total.
   b) Display the **total calories consumed so far**.
   c) Display feedback messages based on progress:
      - 0–800 → "You're off to a healthy start!"
      - 801–1600 → "Good progress, keep it balanced!"
      - 1601–1999 → "Almost at your limit!"
      - 2000+ → "Goal reached! Stay mindful!"
   d) Keep track of the **number of entries** the user has made
   e) If the user logs calories **more than 10 times** without reaching the goal, display **"Be cautious of frequent snacking!"**

Sample Input (User Calorie Input)	Sample Output (Feedback)
300, 500, 700	Good progress, keep it balanced!
1000, 800, 300	Goal reached! Stay mindful!
12 entries, total 1500	Be cautious of frequent snacking!

2. **[10]**

You are managing a **movie night event** where you need to calculate how many screens to rent and how much money will be wasted due to empty seats. Write a PHP function that takes three integer inputs: the total number of people attending, the seating capacity of one screen, and the ticket price per seat. The function should calculate how many **complete screens** are needed to accommodate all people (you cannot book partial screens), and how many **empty seats** will remain after everyone is seated. Furthermore, each screen has a **fixed rental cost of 25,000 BDT**, and the amount of money wasted on empty seats must be calculated based on the ticket price multiplied by the number of unused seats. The output should display the total number of screens booked, total empty seats, and total wasted money.

Sample Input (from HTML fields)				Sample Output (print on PHP file)		
Attendees	**Seat Capacity**	**Ticket Price**		**Total Screens**	**Empty Seats**	**Wasted Money**
150	60	500		3	30	15,000
280	100	400		3	20	8,000
320	120	350		3	40	14,000

3. Create a database named **'sundarban'** and a table named **'sales_data'** with the following structure and values: **[10]**

SaleID	ProductName	CategoryID	CategoryName	Quantity	Revenue
1	Laptop	301	Electronics	5	350000
2	Mouse	301	Electronics	15	45000
3	Chair	302	Furniture	8	64000
4	Desk	302	Furniture	6	72000
5	Bottle	303	Accessories	20	30000
6	Pen	303	Accessories	25	20000

Now, do the following (write full PHP–MySQL code):

1. Display the **total revenue per category** from the database.
2. If a product's revenue is below **40,000 BDT**, update its category to **"Low Performing"**.
3. For all products generating more than **70,000 BDT**, add a **10% bonus revenue** to their total.
4. Display each **product's name** along with its **category name** and a label "**Top Seller**" if its revenue is above the average revenue of its own category. Otherwise, display "**Regular Seller**".

United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2025 Trimester: Fall

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Write a PHP program to design a web page that processes a student's CT and exam marks. The page should take input from an HTML form for three CT marks, a midterm mark, and a final exam mark. The program must select the best two CT marks out of the three, calculate their average, and then compute the total marks by adding the CT average with the midterm and final exam marks. Finally, display the status as 'Passed' if the total marks > 54 otherwise display 'Failed'. For better understanding, refer to the demo input and output. **[10]**

Sample Input (html)	**Sample Output (php)**
CT 1: 12 CT 2: 15 CT 3: 14 Midterm Marks: 22 Final Marks: 32 Calculate Total	Best two CT's total: 29 Midterm marks: 22 Final marks: 32 **Total Marks = 83** Status: Passed

2. Write a PHP program for a simple event-planning system. The program should take three inputs from the user: **number of items sold per day**, **number of days the sale runs**, and **target number of items to be sold**. First, calculate the **total items sold** by multiplying the items sold per day with the number of days. Classify the sales performance using the following conditions: **[10]**

- **Excellent:** total items sold 500 or more
- **Good:** total items sold 300 or more but less than 500
- **Average:** total items sold 150 or more but less than 300
- **Poor:** total items sold less than 150

The program should then compare the total items sold with the target number of items. Finally, display the total items sold, the performance category, the target value, and clearly state whether the sales are **above**, **below**, or **exactly at** the target, along with the difference.

Sample Input (taken using html form)			Sample Output (printed using PHP code)		
Items Sold	Number of	Target	Total Items	Performanc	Result
25	12	300	300	Good	Target met exactly (0
40	10	350	400	Excellent	Above target by 50
15	10	200	150	Average	Below target by 50
8	10	120	80	Poor	Below target by 40

3. Create a database named **campus_library** and a table named **book_loans**. The table tracks student borrowing habits and book conditions. **[10]**

LoanID	StudentName	BookTitle	DaysOverdue	PenaltyFee	Status
101	Abdul	Data Structures	0	0.00	Returned
102	Jabbar	Operating Systems	12	24.00	Overdue
103	Barkat	Discrete Math	5	10.00	Overdue
104	Rahim	Linear Algebra	2	4.00	Overdue
105	Karim	Data Structures	15	30.00	Lost
106	Fahim	Operating Systems	0	0.00	Returned

Answer the following queries

1. Show the total number of books for each Status (*Returned*, *Overdue*, *Lost*), but only include statuses that have more than 1 entry in the table.
2. For any student who has a Status of '*Overdue*' and DaysOverdue less than 7, change their Status to '*Grace Period*' and set their PenaltyFee to 0.
3. For students with a PenaltyFee greater than 20.00, increase their fee by 10% as a "*processing charge*," but only if the final fee does not exceed 50.00.
4. Display each BookTitle and the total PenaltyFee collected/owed for that book. Sort the results so that the book generating the most money appears at the top.

United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2026 Trimester: Spring

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

SET - A

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Develop a JavaScript application where users can track their weekly study hours by subject and get feedback. **[10]**

At first, create two input fields: **Subject Name** and **Study Hours**, an **"Add Study"** button, and a label area to display total study hours, average study hours, subjects studied, and feedback. The user will enter the subject name and the number of hours studied in each session. When the button is clicked, the application will do the following:

a. Set a weekly study goal of **20 hours**.
b. Add the entered study hours to a running total.
c. Store each entered subject name.
d. Count how many study sessions the user has entered.
e. Calculate the average study hour so far:
   Average = Total Study Hours / Number of Sessions
f. Display:
   - total study hours
   - average study hours
   - subjects studied in all sessions
   - feedback message
g. Display feedback based on total study hours:
   - 0–5 → "Start studying regularly!"
   - 6–12 → "Good progress!"
   - 13–19 → "Almost ready!"
   - 20+ → "Weekly goal reached!"
h. If the user enters more than 7 study sessions without reaching 20 hours, display **"Increase your study time!"**

Sample Input	Sample Output
English-2, English-3, Bangla-4, Physics-5	Total: 14 hours, Average: 3.5 hours, Subjects: [English, English, Bangla, Physics], Almost ready!
Math-5, Physics-6, Chemistry-7, English-3	Total: 21 hours, Average: 5.25 hours, Subjects: [Math, Physics, Chemistry, English], Weekly goal reached!
8 sessions, total 15 hours	Total: 15 hours, Average: 1.875 hours, Increase your study time!

2. Write a PHP program for a student academic tracking system. The program should take three inputs from the user via an HTML form: **[10]**

- **credits earned per course**
- **number of courses completed**
- **target graduation credits**.

First, calculate the **total credits earned** by multiplying the credits per course by the number of courses completed. Classify the academic progress using the following conditions:

- **Excellent:** total credits 120 or more
- **Good:** total credits 90 or more but less than 120
- **Average:** total credits 60 or more but less than 90
- **Poor:** total credits less than 60

The program should then compare the total credits earned with the target graduation credits. Finally, display the total credits, the progress category, the target value, and clearly state whether the progress is **above**, **below**, or **exactly at** the target, along with the difference.

Sample:

Credits/ Course	Courses Completed	Target	Total Credits	Progress	Result
4	30	120	120	Excellent	Target met exactly (0 difference)
3	35	100	105	Good	Above target by 5
3	25	90	75	Average	Below target by 15
4	10	60	40	Poor	Below target by 20

3. Create a database named tourism and a table named tourist_spot with the following structure and values: **[10]**

SpotID	SpotName	Region	Category	Rating	EntryFee	VisitorsPerYear
1	Cox's Bazar Beach	CTG	Beach	4.9	0	150000
2	Sundarbans Forest	KHL	Nature	4.3	150	80000
3	Ratargul Forest	SYL	Nature	4.8	100	60000
4	Kuakata Beach	PTU	Beach	4.5	0	90000
5	Shat Gambuj Mosque	BGH	Heritage	4.7	30	70000
6	Paharpur	NWG	Heritage	4.1	30	40000

Now, do the following **(write full PHP–MySQL code):**

1. Display the SpotName, Region and Rating of all spots whose rating is above 4.5 or whose SpotName contains the word 'Beach'. Sort the result by Rating in descending order.
2. Update: for all spots whose rating is between 4.0 and 4.5 (inclusive) and whose EntryFee is greater than 0, increase the Rating by 0.2 and raise the EntryFee by 10%.
3. For each category, show the category name, the number of spots in it, the average rating, and the total visitors per year but only include categories whose average rating is above 4.4. Sort the result by average rating in descending order.
4. For each region, show the region name and its total estimated yearly revenue, calculated as the sum of (EntryFee × VisitorsPerYear) over its spots. Display only those regions whose total estimated revenue exceeds 1,000,000, ordered from highest to lowest.

United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2026 Trimester: Spring

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

SET - B

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. **[10]**

Create a JavaScript program where a user enters a patient's vitals and your code shows a live health report. Strategies are as follows:

Index.html:

Create two text fields (input type text) to take the user's **Heart Rate (bpm)** and **Oxygen Saturation / SpO2 (%)**, and a **"Save Reading"** button. Below the button there is a label field to show the report and feedback.

Script.js:

1. Take the user's Heart Rate and SpO2 from the input fields in the HTML file.
2. Keep track of every reading entered so far (for example, in arrays).
3. Inside a function:
   a) Calculate and display the Average Heart Rate (of all readings entered so far, including the current one).
   b) Calculate and display the Average Oxygen Saturation (of all readings so far, including the current one).
   c) Calculate and display the **Risk Score** from the current reading using the formula:

$$\text{RiskScore} = (100 - \text{SpO2}) \times 2 + |\text{HeartRate} - 80| \times 0.5$$

(|HeartRate − 80| is the absolute / positive difference, e.g. |78 − 80| = 2.)

   d) Display the **feedback** in the label area: "SAFE" if the Risk Score is 10 or less, "WARNING" if it is greater than 10 and 20 or less, and "DANGER" if it is greater than 20.

Sample Input / Output

Sample Input	**Sample Output**	
Heart Rate, SpO2 (entered in order)	Report	Feedback
80, 97	Average Heart Rate: 80 Average Oxygen Saturation: 97 Risk Score: 6.0	**SAFE**
78, 95	Average Heart Rate: 79 Average Oxygen Saturation: 96 Risk Score: 11	**WARNING**
130, 80	Average Heart Rate: 96 Average Oxygen Saturation: 90.67 Risk Score: 65	**DANGER**

2. **[10]**

The UIU Mars Rover Team is preparing for a rover field-test mission and needs to calculate how many battery modules must be carried for the rover. Write a PHP program that takes three inputs from an HTML form: the **mission distance in meters**, the **energy consumption per meter**, and the **capacity of one battery module**.

The program should first calculate the **total energy required** by multiplying the mission distance with the energy consumption per meter. Since the team cannot carry partial battery modules, the program must calculate the minimum number of **complete battery modules** needed. Then, calculate the **unused energy capacity** after the required number of battery modules is carried.

Classify the battery usage efficiency using the following conditions:

- **Efficient:** unused energy is less than or equal to 10% of total carried energy
- **Acceptable:** unused energy is more than 10% but less than or equal to 25% of total carried energy

- **Wasteful:** unused energy is more than 25% of total carried energy

Finally, display the **total energy required**, **number of battery modules needed**, **unused energy capacity**, and the **battery usage status**.

Sample Input taken using HTML form			**Sample Output printed using PHP code**			
Mission Distance	Energy per Meter	Battery Capacity	Total Energy Required	Battery Modules	Unused Energy	Status
1200	8	5000	9600	2	400	Efficient
1500	9	4000	13500	4	2500	Acceptable
1000	6	5000	6000	2	4000	Wasteful

3. Create a database named 'uiu_bookshop' and a table named 'book_info' with the following structure and values: **[10]**

BookID	**BookTitle**	**Author**	**Category**	**Price**	**Stock**
1	Web Basics	John Smith	Programming	450	10
2	Database Guide	Alice Roy	Programming	550	8
3	English Grammar	Mary Khan	Language	300	15
4	Business Math	David Lee	Business	400	12
5	Python Mastery	Sara Ahmed	Programming	600	5
6	French Made Easy	Paul Costa	Language	350	7

Now, do the following **(write full PHP–MySQL code):**

1. Display all books from the Programming category, sorted by Price in descending order.
2. Display the BookTitle, Author, and Price of all books where the Price is greater than 400 BDT and the Category is 'Programming'.
3. Display the number of books available in each Category.
4. Display the total net worth of the entire bookshop. (Hint: To find the total net worth of the entire bookshop, multiply the Price and Stock of each book, then add all the values together.)
5. Apply a 10% price reduction to all books in the 'Language' category, then display the BookTitle and updated Price of those books.